

TinyML-Based Classroom Occupancy Estimation for Energy-Efficient Building Applications

Development Board: NUCLEO-L476RG

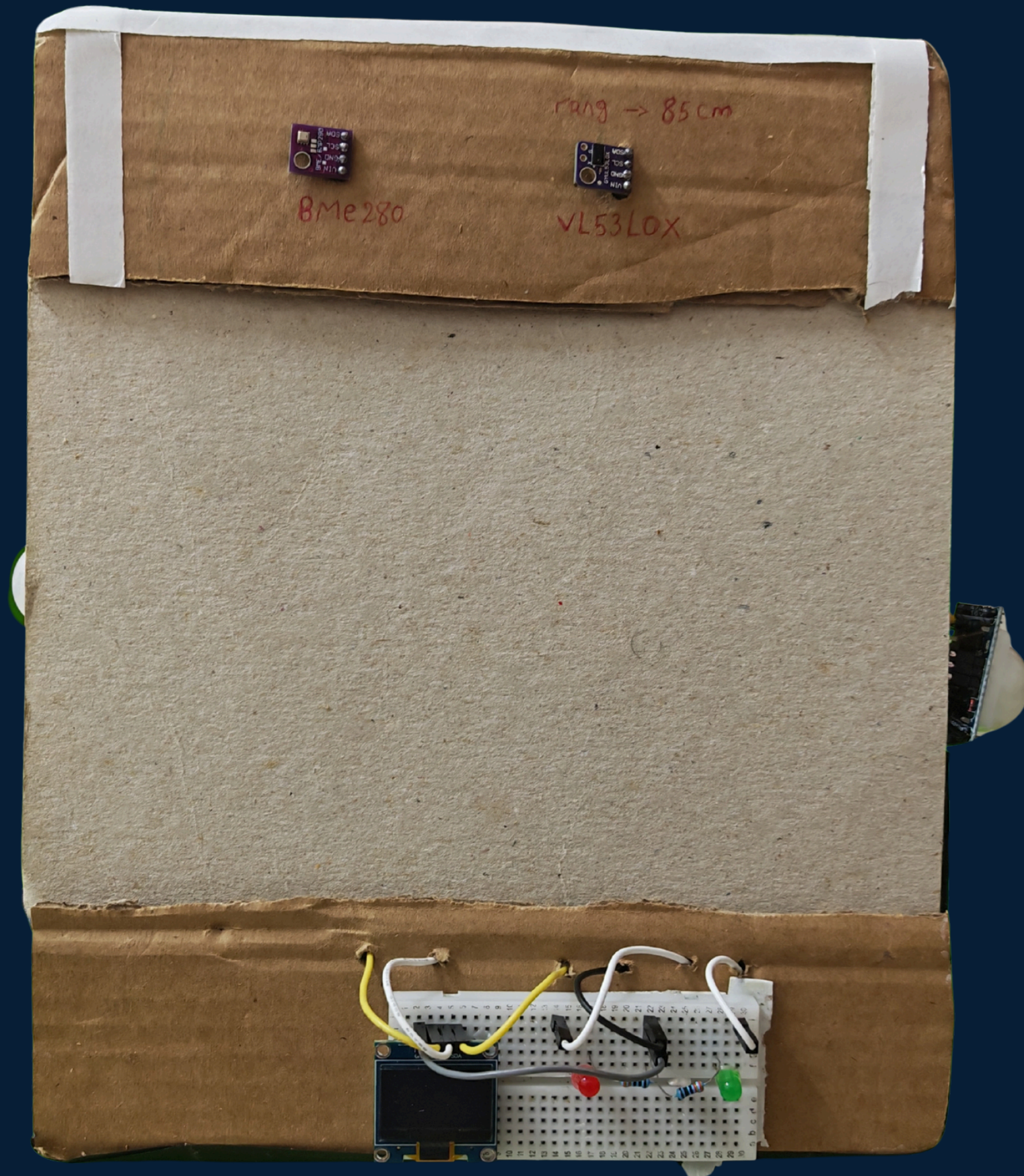
Project Area: TinyML / Edge AI / Smart Building

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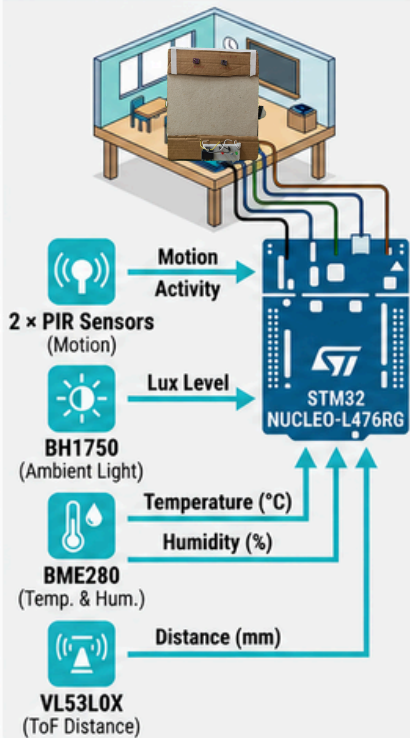
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VeCAD Elite Internship 2026

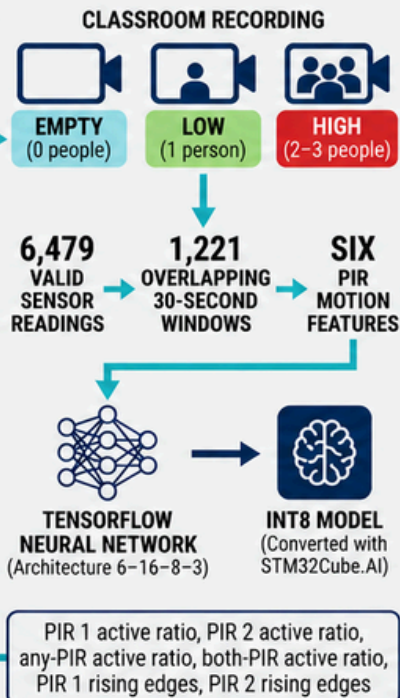


TinyML-Based Classroom Occupancy Estimation for Energy-Efficient Building Applications

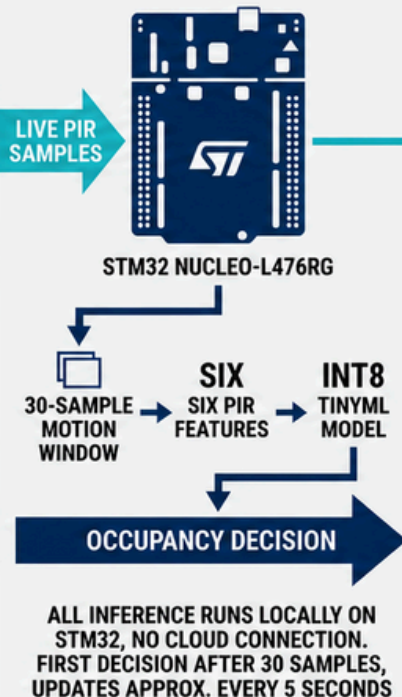
SECTION 1: CLASSROOM SENSING



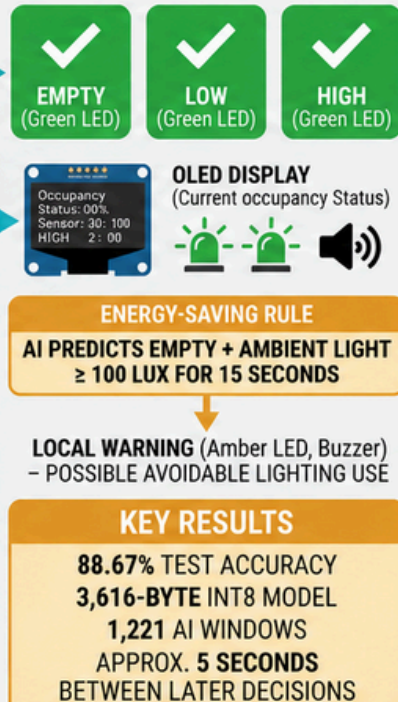
SECTION 2: DATA COLLECTION AND TINYML TRAINING



SECTION 3: LIVE EMBEDDED INFERENCE



SECTION 4: OUTPUT AND ENERGY FEEDBACK



Low-voltage decision-support prototype: it provides a warning and does not switch classroom mains power directly.

Introduction

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Indoor room often consume energy for lighting and air-conditioning and other room equipment consume energy under changing occupancy conditions. even when only a few people are present or when the room is empty. Therefore, occupancy information can help identify opportunities to reduce unnecessary energy use.

- This project develops a low-cost Smart Classroom Occupancy and Energy Monitoring prototype using theSTM32 NUCLEO-L476RG and multiple sensors. The system estimates the classroom occupancy condition in real time as EMPTY, LOW, or HIGH.
- A lightweight TinyML model runs directly on theSTM32 microcontroller. The estimated occupancy state is then displayed locally and used with environmental readings, such as light level, to provide simple energy-saving warnings.

This is a safe low-voltage smart-building prototype.

It provides a decision-support indication; it does not connect directly to room mains power, lighting circuits or air conditioning equipment.



Energy use may continue although room occupancy has changed.

- **Occupancy–energy mismatch:** Lighting and air-conditioning can remain ON after occupants leave a room or when only one person is present. Energy use may therefore exceed the needs of the current room activity.
- **Limited occupancy visibility:** The ON/OFF state of room equipment does not show whether the room is EMPTY, LOW, or HIGH. Without current occupancy information, possible avoidable energy use cannot be identified promptly.
- **Embedded deployment constraint:** The solution must estimate occupancy locally within the memory and processing limits of the STM32 microcontroller. It must remain a safe low-voltage decision-support prototype without direct connection to building mains equipment.

From Low-Voltage Sensing to Local Energy-Aware Feedback

Main Research Question

How can a low-cost, low-voltage STM32 system turn indoor sensor readings into a local occupancy estimate and an energy-aware indication?

RQ1 — Low-Voltage Sensor Integration

How can multiple low-cost, low-voltage sensors be integrated with the STM32 to capture indoor activity and room conditions safely?

RQ2 — Data Collection and Labelling

How can the collected sensor readings be organised into reliable labelled sessions for EMPTY, LOW, and HIGH occupancy conditions?

RQ3 — Occupancy Prediction from Motion

How can the two PIR motion signals be converted into time-window features that distinguish the three occupancy conditions?

RQ4 — Embedded TinyML Deployment

How can the trained occupancy classifier be converted into a compact INT8 model and run locally on the STM32 NUCLEO-L476RG?

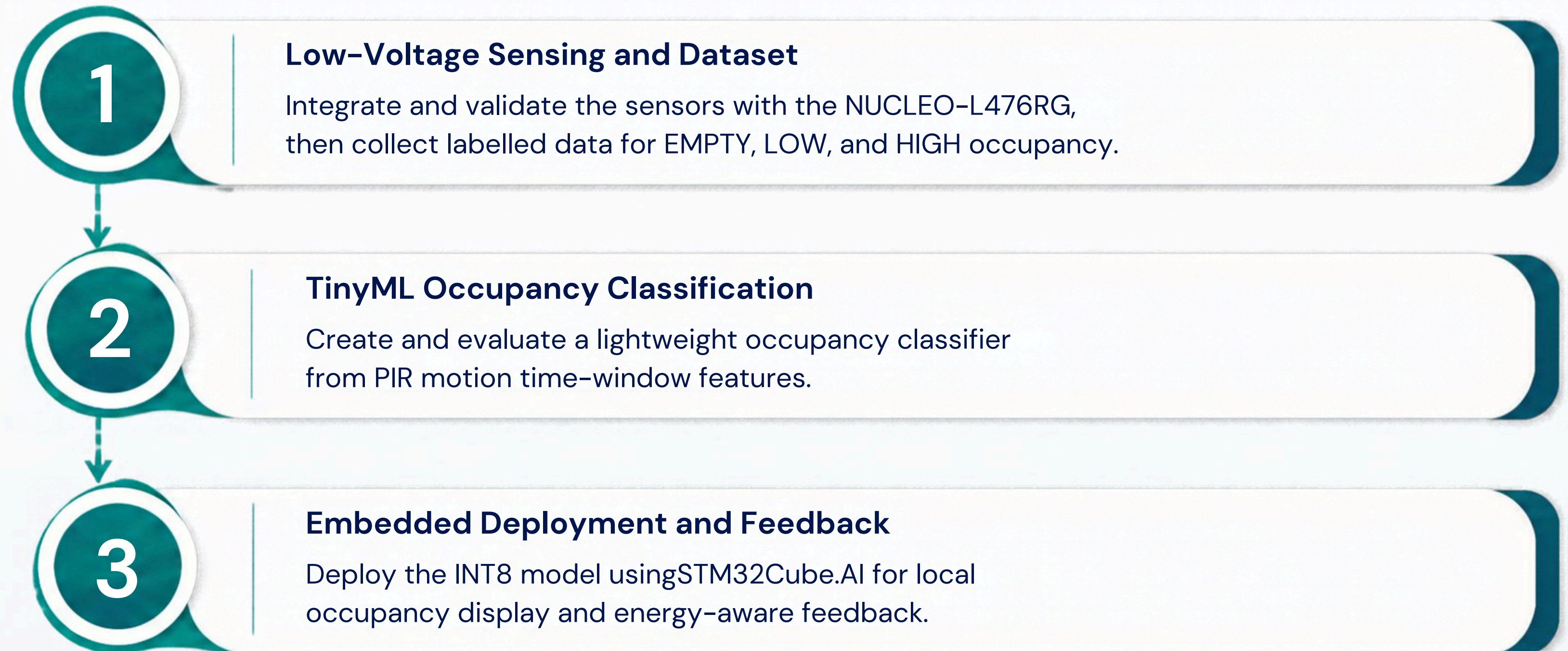
RQ5 — Energy-Aware Feedback

How can the local occupancy result and room-condition information support an energy-aware indication without connecting directly to lighting or air-conditioning circuits?

Project Objectives

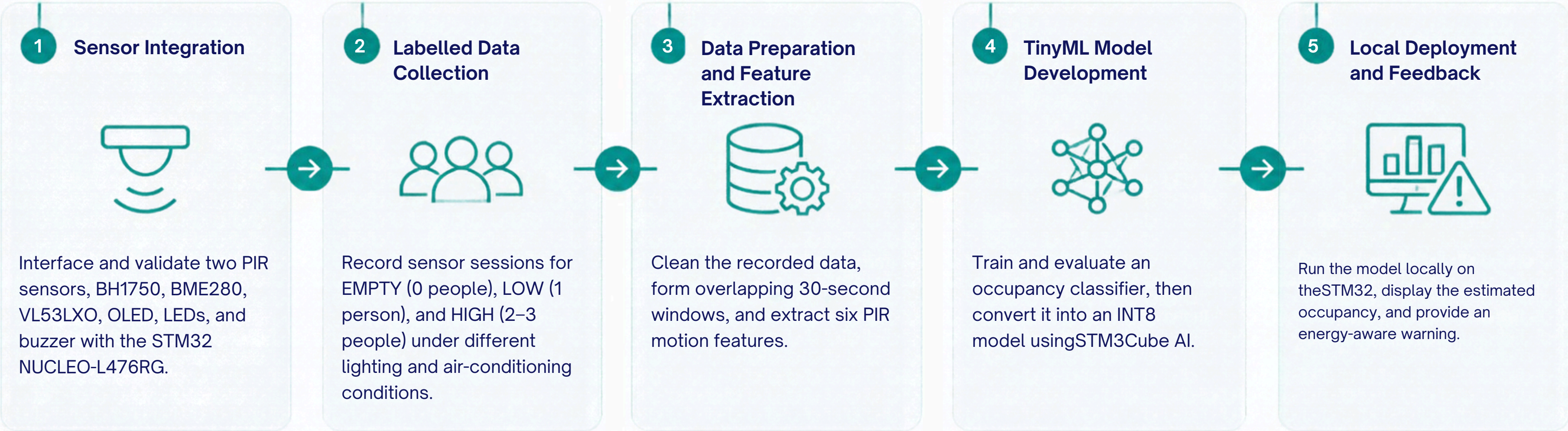
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The project applies a complete TinyML workflow to a safe low-voltage indoor occupancy prototype.



Methodology Overview

One connected workflow converts safe low-voltage sensor readings into a local occupancy estimate and energy-aware feedback.

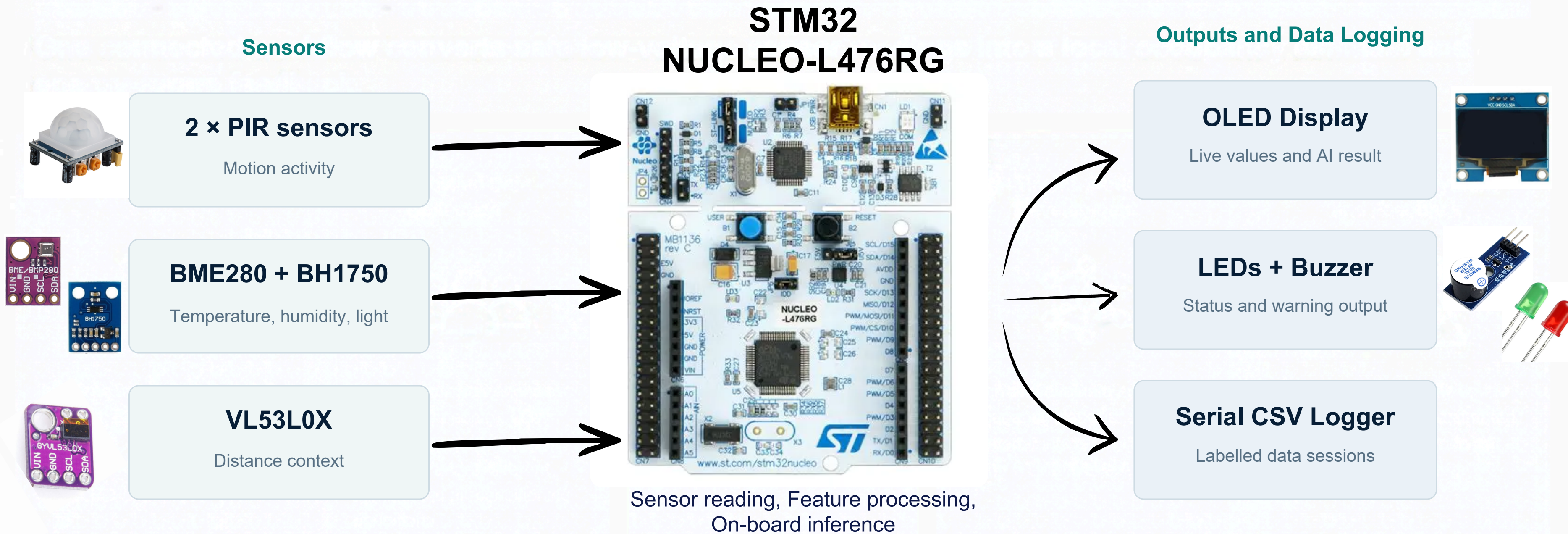


Note:

The AI model uses PIR motion features only. The other sensors provide room monitoring and energy context.

1) System Architecture

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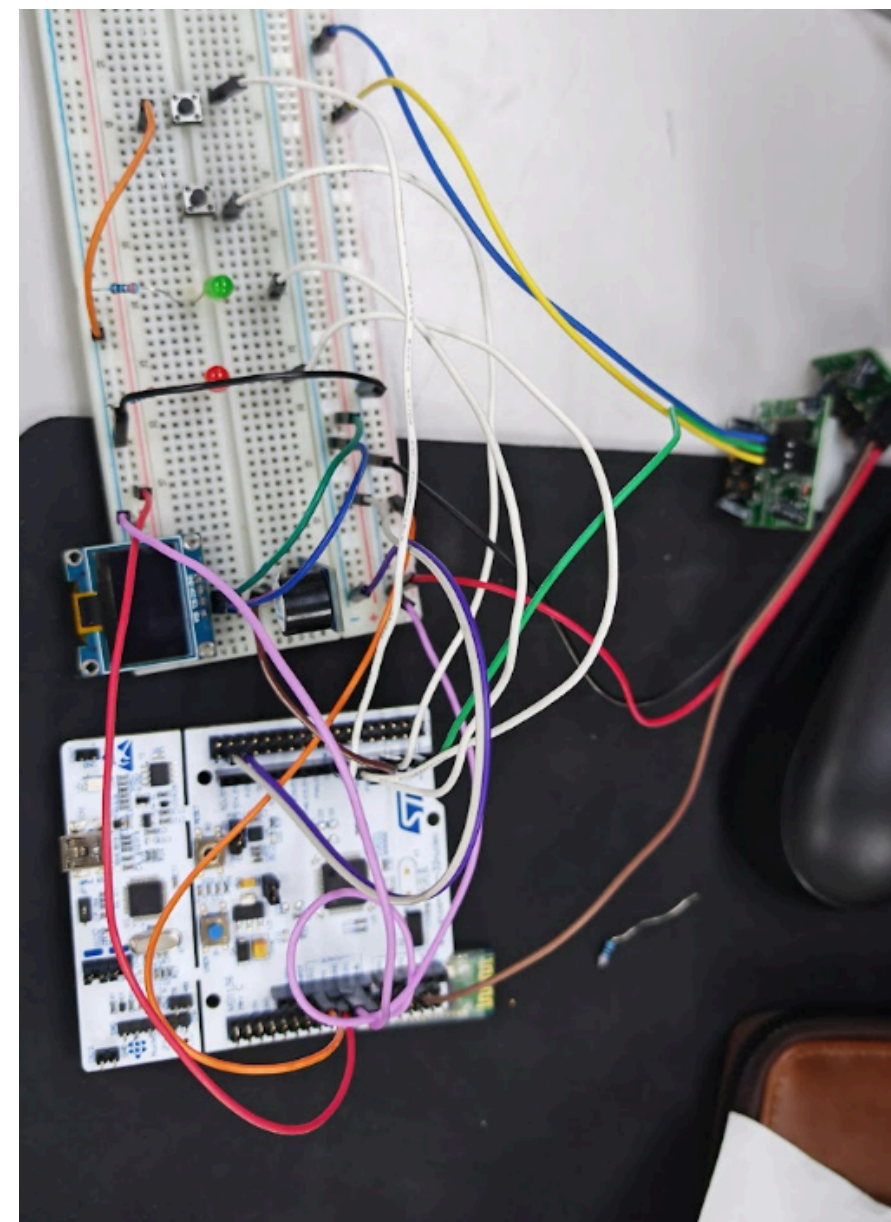
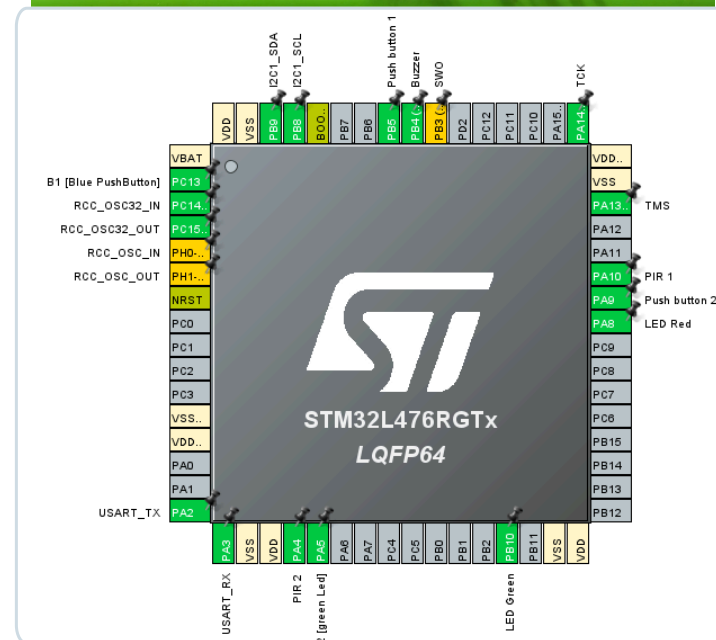
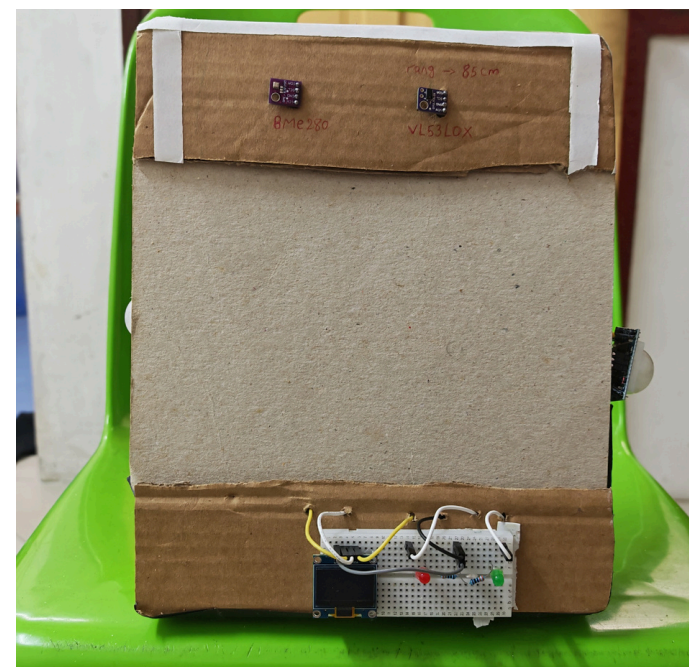


- PIR motion data are used to calculate the six AI features for occupancy classification.
- BME280 provides temperature and humidity monitoring. BH1750 measures ambient light for the energy-warning condition.
- VL53L0X provides short-range distance context and is not used as input to the current AI model because its coverage is limited..

Prototype Portable Occupancy Monitoring Prototype

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The prototype integrates the STM32 board, sensors, control buttons and local feedback outputs in one portable low-voltage enclosure.



- Two PIR sensors provide motion coverage for occupancy estimation.
- I2C sensors and the OLED collect and display room information. Buttons select the occupancy label and control each recording session.

I2C bus: PB8 / PB9

- BME280: temperature, humidity and pressure
 - BH1750: ambient light in lux
 - VL53L0X: distance measurement
 - SSD1306: OLED display output
- LEDs and the buzzer provide local status and warning feedback.

Integrated prototype used for sensor testing, labelled data collection and live demonstration.

2) Data Collection and Labelling



During each recording session, theSTM32 samples the sensors once per second and sends labelled records through UART. A Python logger saves every session as a CSV file.

Button 1 selects the occupancy label: **EMPTY**, **LOW** or **HIGH**.

- Select **EMPTY** label: no person in the room
- Select **LOW** label: 1 person in the room
- Select **HIGH** label: 2-3 people in the room

Button 2 starts and stops the recording session.

- Start a new data-recording session
- Stop the current recording session

Sessions were recorded under different lighting and air-conditioning conditions.

Start and end transition samples were removed before model training.

6,479

valid sensor readings

1,221

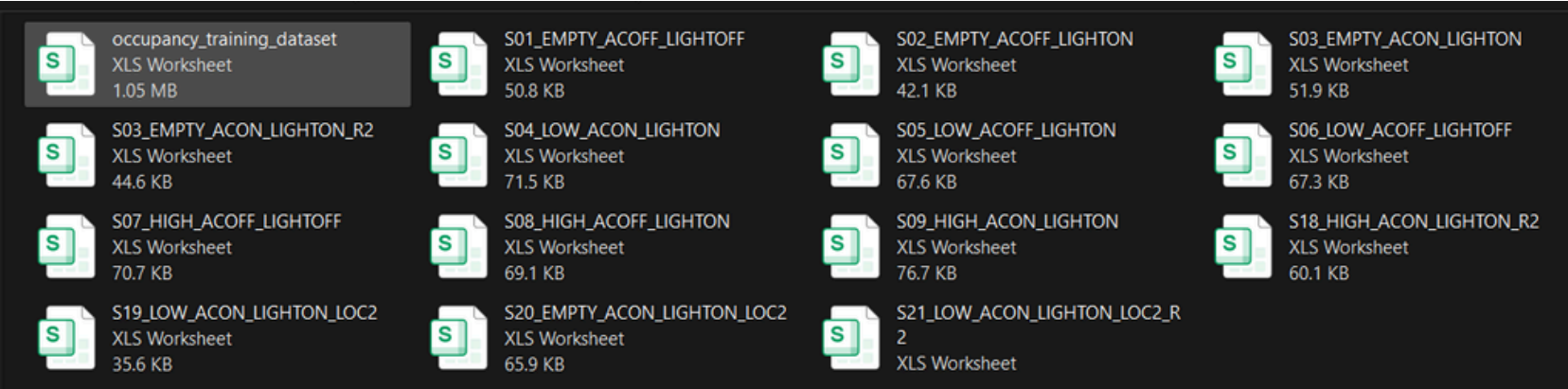
overlapping AI windows

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occupancy classes

Raw CSV field	Unit/type	Meaning	Cleaning rule
pir_1	0/1	PIR sensor 1 state	Only 0 or 1
pir_2	0/1	PIR sensor 2 state	Only 0 or 1
tof_mm	mm	VL53L0X distance	20–2000 mm after status check
tof_status	code	VL53L0X RangeStatus	0 is valid; other values are invalid
temperature_c	°C	BME280 temperature	–40 to 85 °C
humidity_percent	%RH	BME280 humidity	0 to 100%
pressure_hpa	hPa	BME280 pressure	300 to 1100 hPa; retained in raw
light_lux	lux	BH1750 ambient light	0 to 65,535 lux
sample_valid	0/1	BME280 and BH1750 read succeeded	Only rows with 1 are used
occupancy_label	class	Manual ground truth	EMPTY, LOW, or HIGH

EachCSV row keeps the time, PIR data, environmental readings, validity status and selected occupancy label.



Data Collection

3) Dataset Preparation and Quality Control

Raw data is not sent directly to the model. It is checked, cleaned and converted into consistently labelled time windows.

1 Session review

Verify the selected occupancy label and confirm the room condition for each recording.

2 Transition removal

Remove start and end rows that include button presses, entry movement or leaving the test area.

3 Validity filtering

Retain records with usable sensor values and preserve the original session identity.

4 Window generation

Build overlapping 30-sample windows and calculate one set of features per window.

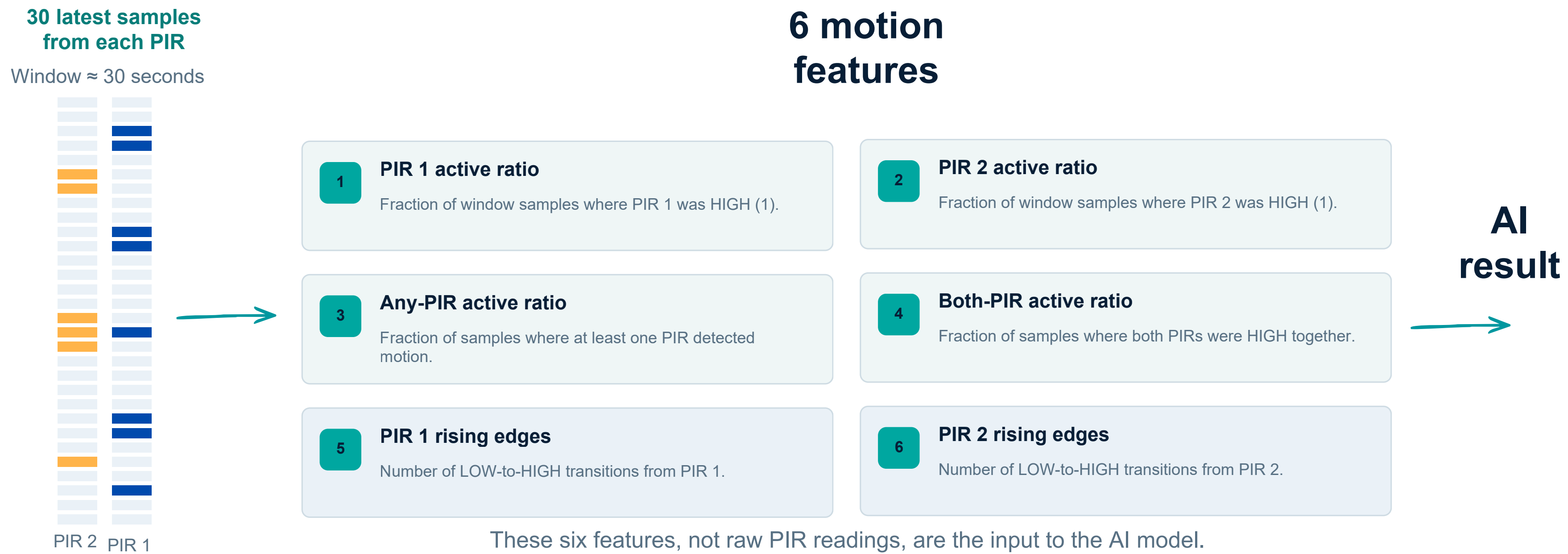
5 Recording-level split

Keep recordings separated across train, validation and test subsets to reduce leakage.

Quality check	Applied rule
Edge trimming	Remove start and end transition readings. The default trim is 5 seconds, with session-specific overrides when needed.
Required records	Keep records with a valid timestamp, numeric readings and sample_valid equal to 1.
Signals and labels	PIR values must be 0 or 1. Sensor readings must be in range. Labels must be EMPTY, LOW or HIGH.
Session integrity	Remove duplicate timestamps and mixed labels. Do not create windows across a gap longer than 5 seconds.
ToF status	A ToF value is valid only when status is 0 and distance is 20 to 2000 mm. Otherwise flag it and encode no target as 2000 mm.

PIR Feature Engineering

The AI model does not use one PIR reading. It summarises a short history of motion from both PIR sensors into six numerical features.



The STM32 forms a new window after every 5 new samples, approximately every 5 seconds.

4) TinyML Training and Evaluation

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Six PIR motion features are standardised, classified by a compact neural network, and evaluated on unseen recordings

Feature Preparation

AI input: six PIR motion features only.

Active ratios range from 0 to 1, while rising-edge counts range from 0 to 29.

Each feature is normalised using the training mean and standard deviation.



Model Training

6 inputs → 16 ReLU neurons → 8 ReLU neurons → 3 Softmax outputs

Outputs: **EMPTY, LOW, HIGH**

- Complete recordings were kept separate across the training, validation, and test sets, so the final test set contained unseen sessions.
- Training settings: Adam optimiser, learning rate 0.001, batch size 32, sparse categorical cross-entropy loss, and class weighting.
- Early stopping selected the best model at epoch 49 from a maximum of 300 epochs.

Training

659 windows

Used to learn model weights. Class weighting reduces the effect of uneven class counts.

Validation

306 windows

Observed during training to choose the best model point and prevent overfitting.

Test

256 windows

Held back until the end for an unbiased final accuracy and F1 measurement.

5) STM32Cube.AI Deployment

The trained model was converted from an INT8 TensorFlow model into embedded C code for the NUCLEO-L476RG.

Deployment path

Keras model→ INT8 TensorFlow Lite → STM32Cube.AI analysis → Generated C network → STM32CubeIDE integration

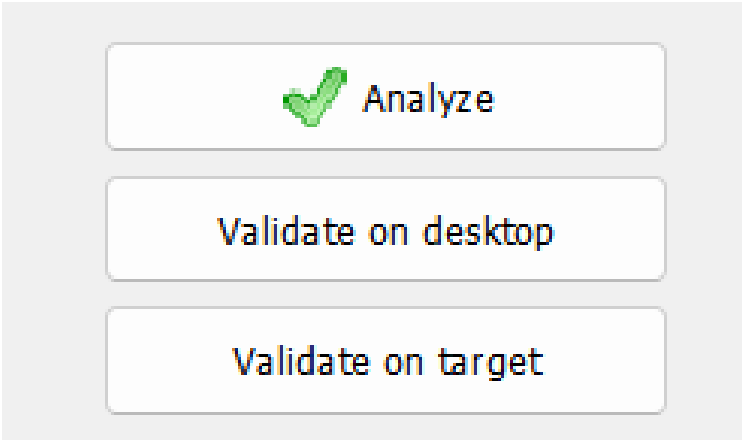
320 MACC operations per inference	13.60 KiB Flash 1.33% of 1 MB internal Flash	2.91 KiB RAM 2.27% of 128 KB internal RAM	3,616 bytes INT8 TensorFlow Lite model
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- The memory and computation requirements are small enough for the NUCLEO-L476RG, leaving substantial resources for sensor, display, and feedback firmware.
- Flash and RAM values refer to the STM32Cube.AI network analysis; the complete firmware also includes sensor and display drivers

```
Elapsed time (analyze): 72.520s
Model file:      occupancy_model_int8.tflite
Total Flash:    13928 B (13.60 KiB)
  Weights:      356 B
  Library:      13572 B (13.25 KiB)
Total Ram:      2976 B (2.91 KiB)
  Activations: 1028 B (1.00 KiB)
  Library:      1948 B (1.90 KiB)
  Input:        6 B (included in Activations)
  Output:       3 B (included in Activations)
Done
Analyze complete on AI model
```

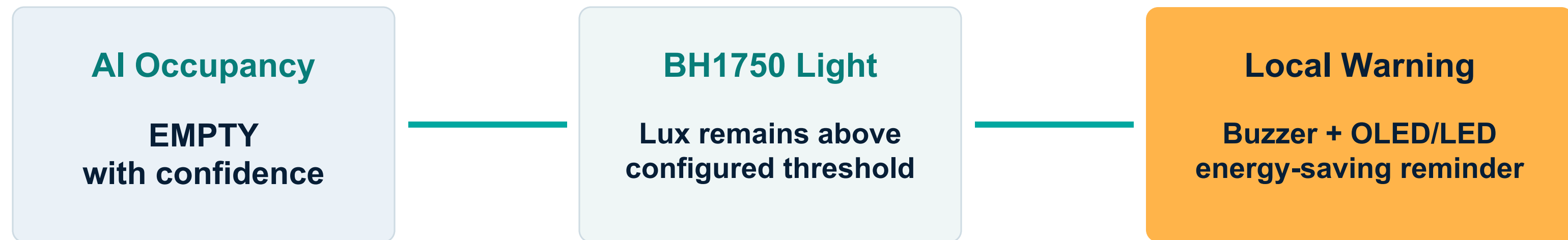
Local Runtime Flow

30 PIR samples → Six PIR features → Same feature normalisation → INT8 model inference → EMPTY / LOW / HIGH result → OLED, LEDs, and buzzer



Energy-Aware Feedback Logic

The AI model estimates occupancy. Rule-based logic uses that result together with environmental sensors to create understandable warnings.



A sustained condition is required before a warning. This avoids reacting to a single momentary light reading or one uncertain AI result.

BME280 temperature and humidity readings can support additional comfort or energy-status alerts, but they are not presented as direct evidence of the number of people.

Results and Discussion

The final model was tested on 256 motion windows from recording sessions that the model did not see during training.

88.67%

test accuracy

227 of 256 unseen windows correctly classified

87.47%

macro F1 score

Balanced precision and recall across the three classes

3,616 B

INT8 model file

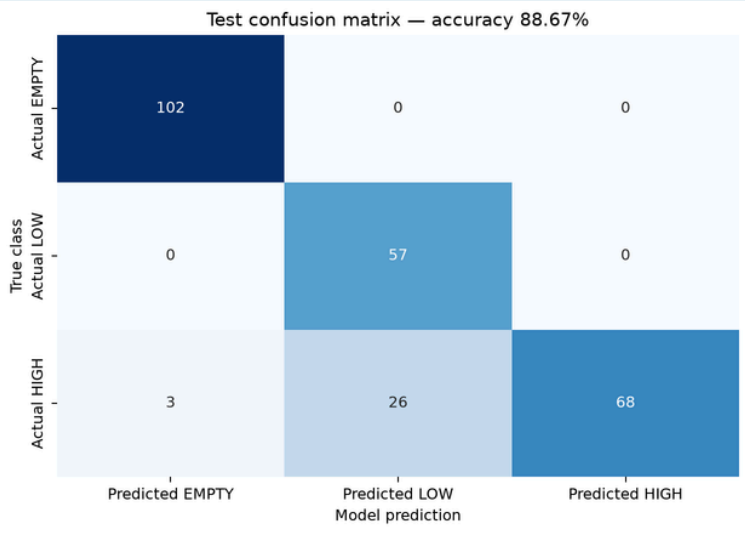
compact Torrentflow Lite model

320 MACC

per inference

small embedded compute demand

Test-set classification



EMPTY: 102/102 correctly classified

LOW: 57/57 correctly classified

HIGH: 68/97 correctly classified

The model identified EMPTY reliably and detected all actual LOW windows. Most errors occurred when HIGH occupancy was predicted as LOW (26 windows),

Embedded deployment analysis

13.60 KiB Flash

1.33% of 1 MB internal Flash

2.91 KiB RAM

2.27% of 128 KB internal RAM

STM32Cube.AI confirmed that the network fits comfortably on the NUCLEO-L476RG, leaving resources for sensing, display output and local feedback.

Flash and RAM values refer to the generated AI network analysis.

Operational result : **EMPTY predicted + ambient light $\geq$ 100 lux for $\geq$ 15 seconds triggers a local OLED, LED and buzzer reminder.**

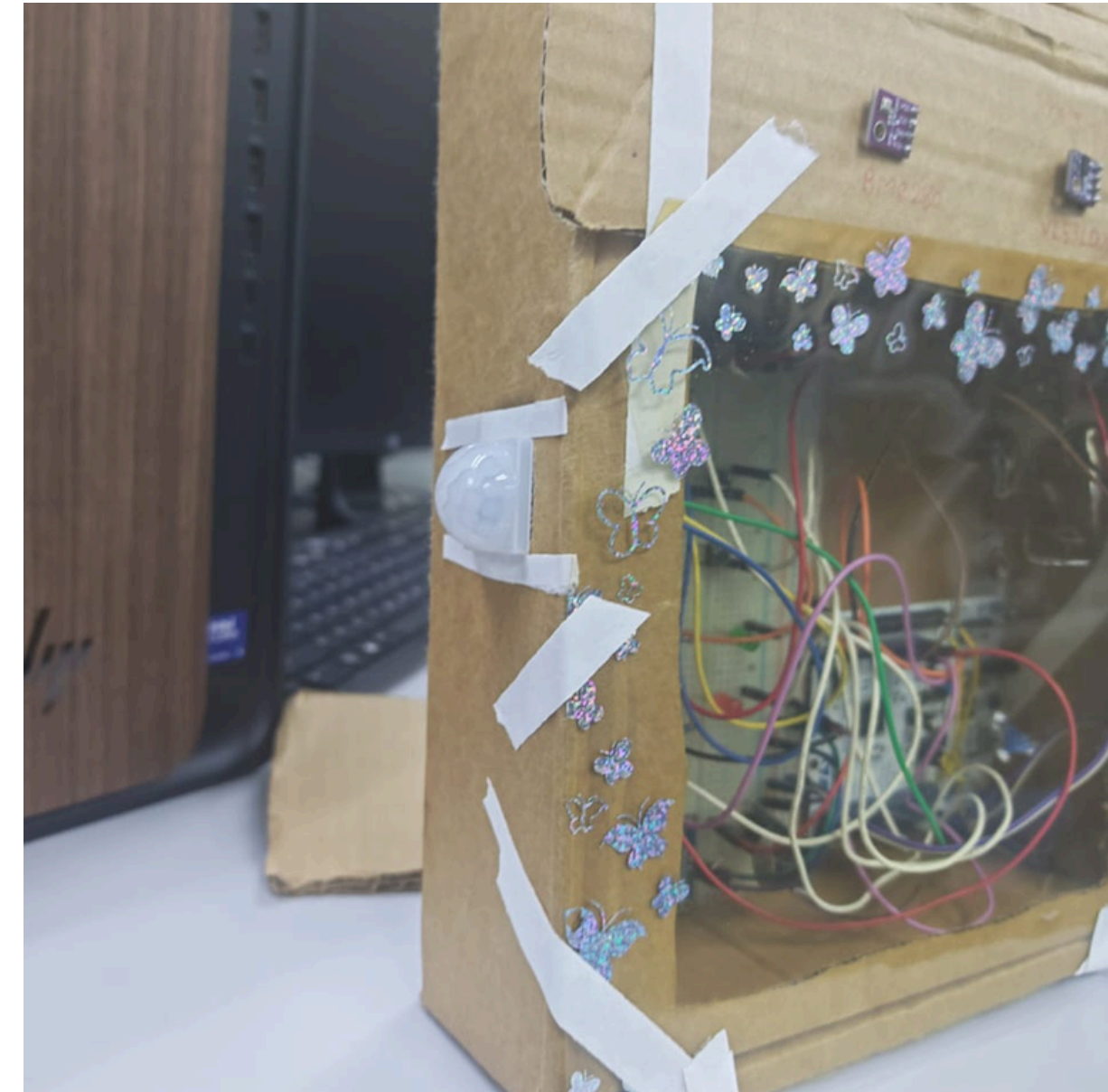
The prototype prioritises low-cost, low-voltage and locally understandable occupancy indication. It estimates occupancy classes rather than counting people exactly.

Engineering decisions

- Two PIR sensors observe motion from different directions and provide a richer activity pattern than one sensor.
- The classifier uses six PIR window features because labelled testing showed that motion activity provided the most consistent occupancy evidence.
- Bh1750 and BME280 readings support energy-aware rules. VL53L0X remains distance context because its narrow view cannot represent the full room.

Observed limitations

- PIR sensors measure movement. A person who remains very still may resemble an EMPTY condition.
- The HIGH class produced varied movement patterns and was sometimes predicted as LOW.
- VL53L0X gives useful close distance data but its narrow field of view and observed range do not cover the full room
- The prototype provides decision support only and never controls building mains, lighting or air-conditioning circuits directly



Integrated low-voltage prototype

Sensor placement and the enclosure influence motion coverage, so room layout remains part of the engineering design

Conclusion and Future Work

The project demonstrated a low-cost, low-voltage STM32 prototype that estimates classroom occupancy locally and provides energy-aware feedback. Its 88.67% test accuracy establishes a validated baseline for further development.

Third PIR coverage

Add a third PIR sensor to reduce blind areas, then collect new EMPTY, LOW and HIGH recordings

Retrain and compare

Train a three-PIR model and compare its accuracy, macro F1 score and HIGH-occupancy performance with the current baseline.

Sensor fusion evaluation

Evaluate distance and environmental features only when new data proves that they improve generalisation.

Practical enclosure

Move from the cardboard prototype to a neat fixed enclosure with safer cable management and permanent mounting.

References

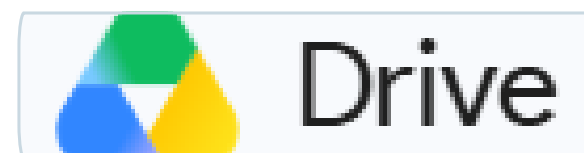
Technical documentation, software resources and project evidence

Embedded platform and software

- 1 STMicroelectronics. STM32 Nucleo-64 boards (MB1136): User Manual UM1724.
- 2 STMicroelectronics. STM32L476xx ultra-low-power ARM Cortex-M4 microcontrollers: Datasheet.
- 3 STMicroelectronics. STM32Cube.AI expansion package for artificial intelligence: User Manual UM2526.
- 4 TensorFlow. TensorFlow Lite Converter Documentation.

Sensors and project material

- 5 Bosch Sensortec. BME280 combined humidity and pressure sensor: Datasheet.
- 6 ROHM Semiconductor. BH1750FVI digital ambient light sensor: Datasheet.
- 7 STMicroelectronics. VL53L0X time-of-flight ranging sensor: Datasheet.
- 8 Ahmed Mohamed Abdalla Ahmed. Labelled occupancy recordings, TinyML training artefacts, STM32 firmware and STM32Cube.AI analysis, 2026.



**LIVE
DEMO**

THANKS YOU